

Electrodynamics Lab One

Report

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Abstract:

This lab is a fun little experiment in measuring coupling constants, mutual inductances, and self-inductances. It provides insights how orientation and proximity plays a role in electrodynamic component relationships.

I. The Lab:

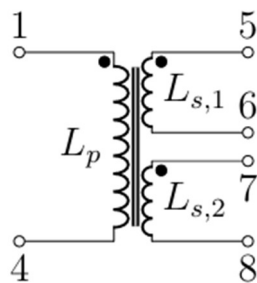
Equipment List & Setup:

ITEM	NEEDED
Ruler or Calipers	1
Alligator clip wires	assortment
Breadboard and leads	assortment
4 Ohm Triad transformer	1
identical 10mH inductors	2
Digital Multimeter (DMM)	1

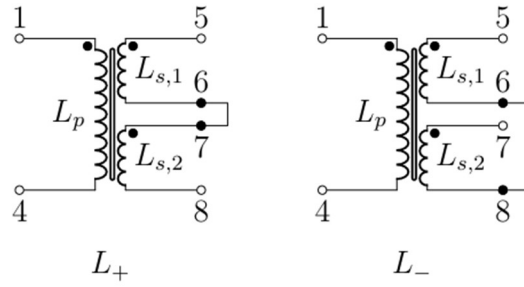
The theory section in the lab instructions provides great insight as to how these inductors are supposed to work in our experiment. I am glad there is finally mention of flux linkage. There was a homework problem that I did using it, but it was just too simple so I took the long route and got the same answer.

This lab asks us to setup multiple different configurations of inductors next to a transformer. These include:

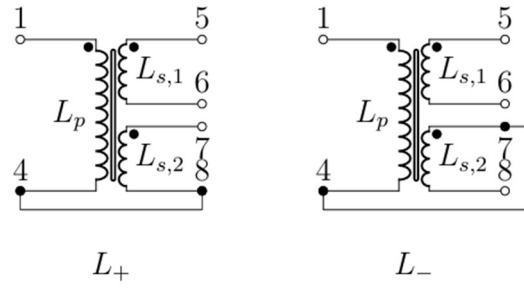
Core Coupling



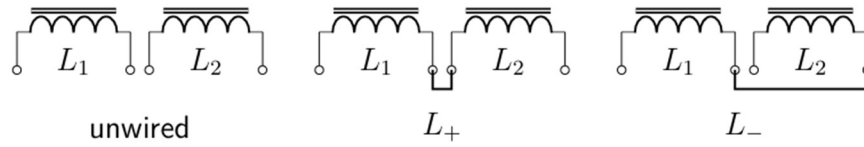
Secondary-Secondary Coupling:



Primary-Secondary Coupling:



Air-Gap Coupling:



II. The Result:

A. Core Coupling:

Inductance Label Measured Inductance

Lp	3.46
L1	0.08
L2	0.08

B. Secondary-Secondary Coupling:

Inductance Label Measured Inductance

L+	0.314
L-	0.004
M	0.078
k	0.969

C. Primary-Secondary Coupling:

Inductance Label Measured Inductance

L+	2.73
L-	3.95
M	0.305
k	3.81

D. Air-Gap Coupling:

Inductance Label Measured Inductance

L1	0.013
L2	0.013
L+	0.001
L-	0.001
M	0
k	0

III. Discussion:

Okay the results of this did not make much sense. When I went back and tried to measure the inductance of each inductor again I was not able to get the DDM to work right. It would measure voltage, resistance, and even capacitance correctly, but I just couldn't get it to work.

One thing I learned in this lab is to never trust lab mates to collect data for you. Collect your own data no matter what. That will be hard to do in the real world so I'll have to trust certain people. The results didn't make sense to me, but the good thing is the lab write up has great information on inductors.